FACE Prep

Simple Sieve

The Simple Sieve is a basic algorithm for finding all prime numbers up to a given limit. Here are the steps:

1. Create a list of consecutive integers from 2 through the limit.
2. Set a variable p to 2, the smallest prime number.
3. Cross out all multiples of p from the list, starting from p^2.
4. Find the next number in the list that is not crossed out. This is the next prime number.
5. Set p to the next prime number found in step 4 and repeat steps 3-5 until p^2 is greater than the limit.
6. The remaining numbers in the list are all prime numbers.

Here's an example of the Simple Sieve algorithm for finding all prime numbers up to 20:

1. Create a list of consecutive integers from 2 through 20: 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20.
2. Set p to 2.
3. Cross out all multiples of p starting from p^2 = 4: 4, 6, 8, 10, 12, 14, 16, 18, 20.
4. The next number in the list that is not crossed out is 3, which is the next prime number.
5. Set p to 3.
6. Cross out all multiples of p starting from p^2 = 9: 9, 12, 15, 18.
7. The next number in the list that is not crossed out is 5, which is the next prime number.
8. Set p to 5.
9. Cross out all multiples of p starting from p^2 = 25. There are no multiples of 5 left in the list.
10. Since p^2 (25) is greater than the limit (20), the algorithm terminates.
11. The rem

```
public class SieveMain {
    public static void simpleSieve(int limit) {
            boolean[] prime = new boolean[limit + 1];
        for (int i = 2; i <= limit; i++) {
            prime[i] = true;
        }

        // Mark all the multiples of the prime numbers
        for (int p = 2; p * p <= limit; p++) {
            // If prime[p] is not changed, then it is a prime
            if (prime[p] == true) {
                // Update all multiples of p
                for (int i = p * p; i <= limit; i += p) {
```

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                prime[i] = false;
            }
        }
    }

    // Print all prime numbers
    for (int p = 2; p <= limit; p++) {
        if (prime[p] == true) {
            System.out.print(p + " ");
        }
    }
}
public static void main(String[] args) {
    simpleSieve(50);
}
}
```

THANK YOU